Nine-Square Arithmetic-Derivative Tuples: An Extension of OEIS A249110 and Their Scaling Structure

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Abstract

OEIS A249110 lists the integers whose squares, together with the squares of their first eight arithmetic derivatives, sum to an integer square. Starting from the accepted five-term prefix, a segmented C computation classifies $823544 \le n \le 22000000000$ and establishes four consecutive terms, ending at $a(9) = 10361580899$. The exclusion claim depends on the AI code-reviewed producer, its finite-domain primality premises, the complete records, and the inherited baseline; it is not supported by a second independent enumeration of all negatives. All nine positive witnesses were independently reconstructed. The resulting next-term bound is $22000000000 < a(10) \le 285311670611 = 11^{11}$. The nine ordered ten-coordinate witnesses have five primitive forms. Classical logarithmic-derivative theory and a product-rule induction give the complete positive integer scales of each observed form: two classes are infinite and three are singletons. Besides the known stationary family p^p, the nonconstant form shared by $a(7)$ and $a(8)$ yields $330004229p^p$ for every prime p. A previously known terminating witness acquires rank six in this sequence; its zero tail explains persistence across later depths. The scale results classify the represented forms, not all possible forms, and no density model is inferred from nine terms.

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1 Definition, prior knowledge and the extension

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$ and let $\mathbb{P}$ be the primes. The ordinary arithmetic derivative on $\mathbb{N}$ is defined by

$$\mathrm{D}(0) = \mathrm{D}(1) = 0, \qquad \mathrm{D}(n) = \sum_{p^e \| n} e\frac{n}{p} \quad (n \ge 2). \tag{1}$$

Here $p^e \parallel n$ means that p^e is the exact power of p dividing n. Equivalently, $\mathrm{D}(p) = 1$ for primes and

$$\mathrm{D}(ab) = a\,\mathrm{D}(b) + b\,\mathrm{D}(a) \qquad (a, b \in \mathbb{N}). \tag{2}$$

These definitions and their logarithmic formulation are classical [2, 3, 4]. Superscripts on D denote composition, with $\mathrm{D}^0(n) = n$. The map is not additive: $\mathrm{D}(2 + 3) = 1$, whereas $\mathrm{D}(2) + \mathrm{D}(3) = 2$.

Definition 1.1. For integers $n, h \geq 0$, put

$$Q_h(n) = \sum_{j=0}^{h} (\mathrm{D}^j(n))^2.$$

For $h \geq 1$, let $S_h = \{n \in \mathbb{N} : n \geq 2,\ Q_h(n) \text{ is a square}\}$. The increasing enumeration of S_8 is denoted $a(1), a(2), \ldots$, with offset one. For $n \in S_8$, define its ordered witness

$$W(n) = (x_0, x_1, \ldots, x_8, R), \qquad x_j = \mathrm{D}^j(n), \quad R > 0, \quad \sum_{j=0}^{8} x_j^2 = R^2. \tag{3}$$

Thus the OEIS title's ten coordinates are the nine chain values and their root; the sequence index is not a coordinate. No primitivity, increasing-leg order, family membership or strict positivity of later iterates is required. Later zero coordinates are admitted. Inputs zero and one are excluded even though the bare square equation has degenerate solutions there. This is the explicit zero-tail convention used throughout.

The sequence is OEIS A249110, contributed by Paolo P. Lava on October 21, 2014 [1]. The snapshot acquired on October 24, 2026, revision 24 dated February 18, 2021, has the five indexed terms

$$4, \quad 27, \quad 3125, \quad 398747, \quad 823543. \tag{4}$$

The entry checked again on October 27 still displays this prefix. Its indexing is inherited here: no new full scan below 823544 was performed. The Maple call printed in the entry is a program, not a retained receipt of an exhaustive search. Its predicate imposes no positivity test on later legs. Moreover, its derivative expression uses the factor list from `ifactors`: the documented lists for zero and one are empty [24], giving zero derivatives. This is a static compatibility check for the zero-tail convention, not a claim to have executed Maple or recovered a historical search receipt.

The subsequence p^p, for $p \in \mathbb{P}$, was credited in A249110 to Ray Chandler on December 22, 2016 [1, 5]. Its mechanism is the known identity $\mathrm{D}(p^p) = p^p$; it is not a discovery of the present computation. Also known before this project was $N = 1072281223$, with a terminating chain and a square sum at every depth at least five, from the author's A249107 study [9]. The first A249110 campaign determines the rank of this known witness, namely $a(6) = N$. It does not discover the integer or its zero-tail mechanism anew.

The first contribution is the finite extension: the subsequent validated prefix of a continuation supplies ranks seven through nine, and a final adjacent run establishes an empty interval ending at 22000000000. Sections 5 and 6 give its method and complete statement after the structural analysis. The mathematical analysis then has two distinct inputs: positive chains, whose identities are individually checkable, and interval coverage, needed to assign consecutive ranks. Neither input substitutes for the other.

The second contribution is the scale structure of these nine witnesses. Their normalization gives five forms, including one nonconstant form represented at scales four and twenty-seven. This relation was recorded while preparing the manuscript handoff, after the numerical results. Here it is extended by proof to an infinite family and placed within the complete scale classification of the five observed forms. The general principles follow logarithmic-derivative theory [3, 4] and the author's earlier fixed-form work [6, 8, 7, 9]; these earlier depth-two through depth-five applications already include nonconstant families. The depth-eight result concerns the particular form displayed here; no absolute priority is claimed.

2 Scaling a chain without an additive derivative

For positive integers s, let

$$\mathrm{L}(s) = \frac{\mathrm{D}(s)}{s} = \sum_{p^e \| s} \frac{e}{p}, \qquad \mathrm{L}(1) = 0. \tag{5}$$

Equation (2) gives $\mathrm{L}(st) = \mathrm{L}(s) + \mathrm{L}(t)$ without a coprimality hypothesis. This is additivity on products, not on sums. For rational $\rho \geq 0$, define

$$T_\rho(m) = \mathrm{D}(m) + \rho m \tag{6}$$

whenever m is a nonnegative integer. An iterate is used only when every preceding value is a nonnegative integer; no derivative of a noninteger is needed in this paper.

Lemma 2.1 (Product-rule induction)**.** *If $t \geq 1$ satisfies $\mathrm{D}(t) = \rho t$ and $m, T_\rho(m), \ldots, T_\rho^j(m)$ are nonnegative integers, then*

$$\mathrm{D}^j(tm) = tT_\rho^j(m). \tag{7}$$

Consequently, when the iterates through eight are integral,

$$Q_8(tm) = t^2 \sum_{j=0}^{8} (T_\rho^j(m))^2. \tag{8}$$

Proof. The claim at $j = 0$ is the identity $tm = tm$. If $z = T_\rho^j(m)$ is the current integer value, the product rule gives $\mathrm{D}(tz) = t\,\mathrm{D}(z) + z\,\mathrm{D}(t) = t(\mathrm{D}(z) + \rho z) = tT_\rho(z)$. This proves the next step, including $z = 0$. Squaring and summing gives (8). No coprimality assumption is used, and no binomial expansion of an additive operator is invoked. □

Definition 2.2. For a witness $W(n)$, let g be the gcd of all ten coordinates, including R. Its primitive ordered form is $F = W(n)/g$. A nonnegative integer tuple $F = (u_0, \ldots, u_8, r)$ with gcd one, $u_0, r > 0$, is called admissible if $sF = W(su_0)$ for at least one positive integer s with $su_0 \geq 2$. Its homothetic class consists of all such witnesses sF.

Normalization preserves order and never includes the index i. An admissible form need not itself be an ordinary derivative chain.

Theorem 2.3 (Criterion for a fixed form)**.** *Let $F = (u_0, \ldots, u_8, r)$ be admissible. Set*

$$\lambda = \frac{u_1 - \mathrm{D}(u_0)}{u_0}. \tag{9}$$

Then $\lambda \geq 0$,

$$u_{j+1} = \mathrm{D}(u_j) + \lambda u_j \ (0 \leq j < 8), \qquad \sum_{j=0}^{8} u_j^2 = r^2. \tag{10}$$

For every positive integer s with $su_0 \geq 2$, $sF = W(su_0)$ if and only if $\mathrm{L}(s) = \lambda$. Every positive integer scale satisfying $\mathrm{L}(s) = \lambda$ automatically has $su_0 \geq 2$. No positive noninteger rational multiple of F has all ten coordinates integral.

Proof. Choose a witnessing scale g. Dividing $\mathrm{D}(gu_0) = gu_1$ by g gives $u_1 = \mathrm{D}(u_0) + u_0\,\mathrm{L}(g)$, so $\lambda = \mathrm{L}(g) \geq 0$. The same argument at every later coordinate gives the transitions, including zeros. Division of the square identity by g^2 gives the last equation. Necessity for any other scale follows from the first transition. Conversely, the product rule and (10) reconstruct all eight derivatives at a scale with $\mathrm{L}(s) = \lambda$, and the square identity scales by s^2. The input cannot be one: if $s = u_0 = 1$ and $\mathrm{L}(s) = \lambda$, then $\lambda = 0$, the transitions force $u_1 = \cdots = u_8 = 0$, and the same rate gives $\mathrm{L}(g) = 0$, hence $g = 1$, since every integer greater than one has a positive term in (5). That would contradict admissibility on inputs at least two. Finally, if a reduced rational a/b makes $(a/b)F$ integral, b divides every coordinate of F, so its gcd forces $b = 1$. □

Ufnarovski and Åhlander's Theorem 18 gives a necessary and sufficient squarefree-denominator condition for nonzero rational solutions of the logarithmic equation and describes all such solutions from a particular one [3, Section 8]. Integer solutions require nonnegative prime exponents. Kovič's Proposition 2 classifies the positive integer solutions at integer rates [4]. The following formulation gives the exact existence criterion and all positive integer solutions at nonnegative rational rates, including the additional condition $k \geq 0$. The same formulation and the rate-1/6 obstruction already appear in the author's earlier fixed-form work [8, Section 3]; they are recalled here for the depth-eight application, without a new general-theorem claim.

Theorem 2.4 (All positive integer scales)**.** *Let $\lambda = A/B \geq 0$ be reduced, $B > 0$. If B is not squarefree, $\mathrm{L}(s) = \lambda$ has no positive integer solution. If B is squarefree, define, for every prime $p \mid B$,*

$$c_p \in \{1, \ldots, p-1\}, \qquad c_p \equiv A(B/p)^{-1} \pmod{p},$$

and put

$$s_0 = \prod_{p \mid B} p^{c_p}, \qquad \alpha_0 = \sum_{p \mid B} \frac{c_p}{p}, \qquad k = \lambda - \alpha_0. \tag{11}$$

For $B = 1$, take $s_0 = 1$ and $\alpha_0 = 0$. Then k is an integer. A positive integer solution exists exactly when $k \geq 0$, and all solutions are

$$s = s_0 \prod_{\ell=1}^{k} p_\ell^{p_\ell}, \qquad p_\ell \in \mathbb{P}. \tag{12}$$

Primes may repeat and may divide B.

Proof. For any $s = \prod p^{e_p}$, the denominator of $\mathrm{L}(s) = \sum e_p/p$ divides a squarefree product; this proves the first assertion. Assume B squarefree. The integer $Bk = A - \sum_{p \mid B} c_p(B/p)$ is divisible by every prime of B, by the defining congruences. It is thus divisible by B, so k is integral, whether or not a solution already exists.

For a solution, let M be the product of the distinct primes dividing sB. Multiplication of the rate equation by M gives

$$\sum_{p \mid s} e_p M/p = AM/B.$$

A prime $p \mid B$ must divide s: otherwise reduction modulo p would make the left side zero and the right side nonzero. Reducing modulo such a p and cancelling the unit M/B gives $e_p \equiv c_p \pmod{p}$. For a prime $p \mid s$ outside B, reduction instead gives $e_p \equiv 0 \pmod{p}$. Therefore $e_p = c_p + ph_p$ on B, and $e_p = ph_p$ outside B, with all $h_p \geq 0$. Substitution gives $k = \sum h_p \geq 0$. Listing a prime p exactly h_p times produces (12). Conversely, every product in that equation has rate $\alpha_0 + k = \lambda$, proving sufficiency and completeness. □

For example, $\lambda = 1/6$ has residue core $s_0 = 2 \cdot 3^2 = 18$, $\alpha_0 = 7/6$, and $k = -1$: its squarefree denominator does not give an integer solution. Rate $1/2$ has only scale two, whereas rate $3/2$ has exactly the scales $2p^p$. In general $k = 0$ gives a singleton scale set; $k \geq 1$ gives infinitely many scales by fixing all but one prime parameter. The rates used below are only zero and one, whose scale sets are respectively $\{1\}$ and $\{p^p : p \in \mathbb{P}\}$. The latter recovers the classical positive fixed-point classification.

3 The five observed forms

Define the two primitive forms

$$\begin{aligned} F_0 &= (1,1,1,1,1,1,1,1,1,3), \\ F_1 &= (330004229, 330099083, 341481839, 361898942, \\ &\qquad 543260957, 560785535, 672942647, 745727049, \\ &\qquad 1286744865, 1933839382). \end{aligned} \tag{13}$$

The subscripts of F_0 and F_1 are labels; both forms have rate one. Only the other three forms carry observed sequence indices as subscripts. They are denoted E_i, distinct from the depth sets S_h:

$$\begin{aligned} E_4 &= (398747, 1692, 2856, 5812, 5816, 8732, 9116, 9500, \\ &\qquad 15700, 399467), \\ E_6 &= (1072281223, 85244735, 27198937, 1431542, 920291, \\ &\qquad 1, 0, 0, 0, 1076009463), \\ E_9 &= (10361580899, 280042764, 373531688, 560459372, \\ &\qquad 611925640, 1040273628, 1387031516, 1515544356, \\ &\qquad 2118671140, 10865566899). \end{aligned} \tag{14}$$

All have gcd one when all ten coordinates are included. Table 1 reconstructs every witness as gF from these five displayed forms. Thus all positive data needed for the mathematical analysis are contained in the paper.

i	$a(i)$	R	g	Form
1	4	12	4	F_0
2	27	81	27	F_0
3	3125	9375	3125	F_0
4	398747	399467	1	E_4
5	823543	2470629	823543	F_0
6	1072281223	1076009463	1	E_6
7	1320016916	7735357528	4	F_1
8	8910114183	52213663314	27	F_1
9	10361580899	10865566899	1	E_9

Table 1: The nine-term local prefix and positive roots. The full ordered witness is the indicated primitive form multiplied by its ten-coordinate gcd g. The first five ranks are inherited; ranks six through nine have the coverage dependencies in Theorem 6.1.

Proposition 3.1. *The complete homothetic classes of F_0 and F_1 have first coordinates p^p and $330004229p^p$, respectively, for $p \in \mathbb{P}$. Each E_i, $i = 4, 6, 9$, has exactly one witness in its homothetic*

class, namely itself. The nine witnesses therefore represent exactly five forms: two infinite classes and three singleton classes.

Proof. For F_0, $\mathrm{D}(1) = 0$ gives rate one, all transitions are $T_1(1) = 1$, and the nine squares sum to $9 = 3^2$. For F_1, let u_j denote its chain coordinates and let $c = u_0 = 330004229 = 3617 \cdot 91237$. The exact derivatives in Table 2 satisfy $u_{j+1} = u_j + \mathrm{D}(u_j)$ for every $0 \le j < 8$, and

$$\sum_{j=0}^{8} u_j^2 = 3739734755374141924 = 1933839382^2. \tag{15}$$

The gcd of the ten coordinates is one. These identities and Lemma 2.1 show that $p^p F_1$ is a witness for every prime p, with no exception when p divides c or another coordinate. Conversely, its first transition forces $\mathrm{L}(s) = 1$, and Theorem 2.4 gives exactly $s = p^p$.

Each E_i is an ordinary derivative chain, and its nine chain coordinates have square sum equal to its last coordinate squared; these finite identities follow by direct reconstruction from (1) and (14). Its first transition gives $\lambda = 0$. Equation (5) then forces $s = 1$, since every integer $s > 1$ has strictly positive logarithmic derivative. The five forms displayed are pairwise distinct; normalizing the nine rows gives precisely the associations of Table 1. □

j	u_j	$\mathrm{D}(u_j)$	u_{j+1}
0	330004229	94854	330099083
1	330099083	11382756	341481839
2	341481839	20417103	361898942
3	361898942	181362015	543260957
4	543260957	17524578	560785535
5	560785535	112157112	672942647
6	672942647	72784402	745727049
7	745727049	541017816	1286744865

Table 2: All eight exact transitions for F_1: the last column is the sum of the middle two columns. These are generalized T_1 transitions, not ordinary derivative transitions of the primitive tuple.

The two observed specializations are

$$330004229 \cdot 2^2 = 1320016916, \qquad 330004229 \cdot 3^3 = 8910114183.$$

The next prime specialization is $330004229 \cdot 5^5 = 1031263215625$. It is an exact unindexed witness outside the scanned interval, and is larger than the historical stationary witness 11^{11}. Hence it does not improve the upper bound for the next rank. Since p^p increases with p, the two observed values exhaust this particular class through the reported frontier. Likewise, $p = 2, 3, 5, 7$ exhaust F_0 there.

The singleton assertion is global for each specified ordered form: no other positive integer scale produces that same form. It does not assert that these integers cannot lie in a parameterized family whose form changes with the parameter. Nor does a complete list of forms in a finite prefix prove that no additional forms exist.

4 Zero tails, depth relations and dynamics

For a fixed point p^p, the sum at depth h is

$$Q_h(p^p) = (h+1)p^{2p}. \tag{16}$$

It is a square exactly when $h+1$ is a square. This explains the roles of four legs at depth three and nine legs at depth eight; fixed points are not members at every depth.

The related OEIS square-sum sequences A230543, A249105, A249106, and A249107 correspond to depths $2, 3, 4, 5$, respectively [10]. A210503 adds primitivity at depth one [11]; the twin-prime construction of primitive triples is also discussed by Balzarotti and Lava [12, Chapter 22]. That restriction is absent from S_8. Individual higher-derivative sequences encode chain coordinates rather than ranks: A258651 is the iterated-derivative array, and A258648 its depth-eight column [13].

Proposition 4.1 (Persistence of the known terminating witness)**.** *Let* $N = 1072281223$ *and* $R = 1076009463$*. Then*

$$\mathrm{D}^5(N) = 1, \qquad \mathrm{D}^j(N) = 0 \ (j \geq 6), \qquad Q_h(N) = R^2 \ (h \geq 5).$$

Moreover $N \notin S_4$*.*

Proof. The exact chain is the nine chain coordinates of E_6 in (14). Here $x_5 = \mathrm{D}^5(N) = 1$, and (1) supplies the zero tail. Its square sum through depth five is $R^2 = 1157796364465548369$; adding zero squares changes nothing. At depth four the sum is $R^2 - 1$, which lies strictly between $(R-1)^2$ and R^2. □

This persistence, including its depth-eight consequence, was already established in the A249107 study [9]. There N has rank twelve in the author's extended A249107 sequence; the public OEIS entry checked on October 27, 2026 still has ten terms and does not include N [10]. Its A249110 rank is six; neither sequence's rank follows from the other. More generally, if $\mathrm{D}^k(n) = 0$, $Q_h(n)$ is constant for $h \geq k-1$, but it need not be square. A zero tail alone is not a membership test.

There is no general inclusion of the sets as depth increases. For example $Q_3(4) = 64$ is square and $Q_4(4) = 80$ is not. In the other direction, the chain beginning $(19164, 25564, 31848, 58412, 61916)$ gives $Q_4(19164) = 96336^2$, while $73804^2 < Q_3(19164) = 5447033840 < 73805^2$. Among the nine inputs of Table 1, the square depths from one through eight are exactly $\{3, 8\}$ for the four stationary inputs, $\{5, 6, 7, 8\}$ for N, and $\{8\}$ for the remaining four inputs. This is a finite correlation check on those nine inputs, not an enumeration at any other depth. In particular, the exclusions of depths one, two and three for N are finite integer-square-root checks; Proposition 4.1 establishes the depth-four exclusion and persistence at all depths at least five.

Proposition 4.2 (Dynamical types in the observed prefix)**.** *The four members of form* F_0 *are stationary,* N *eventually reaches zero, and the four other displayed members have derivatives tending to infinity.*

Proof. The first two assertions follow from fixed points and Proposition 4.1. For F_1, the entire integral T_1 orbit of $c > 1$ satisfies $T_1(m) = m + \mathrm{D}(m) \geq m + 1$ for $m \geq 2$. Lemma 2.1 at every depth therefore shows divergence for all its witnesses. For E_4 and E_9, the first derivatives are respectively 1692 and 280042764, both divisible by four and greater than four. The classical four-divisor argument [3, Theorem 4] can be seen directly: for $m \geq 2$,

$$\mathrm{D}(4m) = 4m + 4\,\mathrm{D}(m) \geq 4m + 4.$$

The next value remains a multiple of four, so induction gives unbounded growth. □

Thus singleton scale classes need not be terminating: two diverge and one terminates in this sample. These pointwise proofs do not resolve the general dynamics of the arithmetic derivative.

5 Exact enumeration and numerical domain

5.1 First-derivative sieve and the remaining chain

The producer is a specialized C17 program with GCC's unsigned 128-bit extension [21], OpenMP, GMP and OpenSSL libcrypto. It supports original inputs through 50000000000, a technical limit distinct from the executed endpoint. The source and build used for the reported runs underwent external AI-assisted code review before those runs, as described in the AI use disclosure. This was neither independent human code review nor a second enumeration of the negatives; manuscript review is a separate stage.

In a block $[L, H]$ of at most 262144 inputs, initialize a residual to each original input n and a derivative accumulator to zero. Generate all primes through $\lfloor\sqrt{H}\rfloor$. For each prime p, visit its multiples, remove its complete exponent e from the residual and add $e(n/p)$, using the original n. A final residual $q > 1$ is prime and contributes n/q. Indeed after all primes through $\sqrt{H}$ have been removed, a composite residual would have a prime divisor at most $\sqrt{H}$, a contradiction. The accumulator is therefore exactly $\mathrm{D}(n)$.

If $\mathrm{D}(n) = 1$, n is prime: a composite $n = ab$, $a, b \geq 2$, has $\mathrm{D}(n) = a\,\mathrm{D}(b) + b\,\mathrm{D}(a) \geq a + b \geq 4$. A prime input has chain $(n, 1, 0, \ldots, 0)$ and $n^2 < Q_8(n) = n^2 + 1 < (n+1)^2$, so this rejection is exact. No family or primitivity restriction is used to remove other inputs.

For the remaining inputs, the producer computes the seven further derivatives by factoring $x_1, \ldots, x_7$ as needed. Small-prime division precedes a fixed-seed rho accelerator using Pollard's quadratic map with Floyd cycle detection [15]. Proper factors recurse on strictly smaller operands. The bounded accelerator uses at most 32 seeds and 100000 steps per seed, then complete odd trial division. Factors are checked as exact divisors, prime leaves are checked, and their product must reconstruct the operand before the derivative is evaluated. The fallback is finite, but no useful uniform practical running-time bound is asserted. Interruption or failure leaves the block incomplete; it never becomes a negative answer. Known zeros are filled with zeros, and an established equality of successive iterates allows the remaining fixed-point coordinates to be filled exactly.

Every coordinate is imported into GMP before squaring. The nine squares are added in multiple precision; the program applies `mpz_perfect_square_p`, extracts the integer root, and checks its square again [20]. It does not square a large coordinate in a narrower native type first.

5.2 Uniform bounds and the two primality domains

A bound for all possible intermediates is needed even when observed positive chains are small. The smallest-prime-factor bound and two-adic decomposition used here are described in [14, Theorems 2.1.1 and 3.2.2], using the classical derivative-bound argument of [3, Theorem 9]. The following integer version makes the rounding explicit. For $U \geq 2$ and $0 \leq a \leq \lfloor \log_2 U \rfloor$, let M_a be the largest odd integer at most $\lfloor U/2^a \rfloor$, and put $k_a = \lfloor \log_3 M_a \rfloor$. Define

$$G(U) = \max_a \left\{ a 2^{a-1} M_a + 2^a \left\lfloor \frac{M_a k_a}{3} \right\rfloor \right\}, \qquad G(0) = G(1) = 0, \tag{17}$$

with the first summand zero for $a = 0$.

Lemma 5.1. *For $0 \leq n \leq U$, $\mathrm{D}(n) \leq G(U)$, and G is nondecreasing.*

Proof. Write $n = 2^a m$, m odd. For $m > 1$, each prime factor counted with multiplicity is at least three, so $\Omega(m) \leq \lfloor \log_3 m \rfloor$ and $\mathrm{D}(m) \leq \lfloor m \lfloor \log_3 m \rfloor / 3 \rfloor$. The inequality also holds for $m = 1$. The product rule gives $\mathrm{D}(n) = a 2^{a-1} m + 2^a\,\mathrm{D}(m)$. For fixed a its upper expression is nondecreasing

in odd m; replace m by M_a and maximize. As U increases each existing M_a is nondecreasing and only additional choices of a enter, proving monotonicity. The cases zero and one are separate. □

Starting at $U_0 = 50000000000$, Lemma 5.1 gives the induction $x_j \le U_j \Rightarrow x_{j+1} = \mathrm{D}(x_j) \le G(U_j) = U_{j+1}$. The resulting chain of uniform bounds is Table 3. The floors can be computed by integer division, without floating-point logarithms.

j	U_j	Bits	Coordinate storage
0	50000000000	36	unsigned 64-bit
1	797790175232	40	unsigned 64-bit
2	14345190768640	44	unsigned 64-bit
3	294669116243968	49	unsigned 64-bit
4	6755399441055744	53	unsigned 64-bit
5	174514485560606720	58	unsigned 64-bit
6	4643211215818981376	63	unsigned 64-bit
7	142962266571249025024	67	unsigned 128-bit
8	4574792530279968800768	72	unsigned 128-bit

Table 3: Universal upper bounds at the supported input cap $5 \cdot 10^{10}$. They are not observed maxima and do not assert coverage to that cap.

The sum bound is 209491864946800109588993692851455160850841 60, requiring 144 bits, and its integer-root bound is 4577028129111728646739, requiring 72 bits. Thus the full arithmetic chain is not confined to 64 bits, and a 128-bit square sum is not justified by these bounds. Even the single-coordinate bound U_8^2 requires 144 bits, so importing the coordinates into GMP must precede multiplication.

The primality predicate is the strong probable-prime test of Miller and Rabin [16, 17], made deterministic on the admitted finite domains by two external premises. For operands below 2^{64}, the bases are

$$2,\ 325,\ 9375,\ 28178,\ 450775,\ 9780504,\ 1795265022,$$

the Jim Sinclair set recorded in [18]. Bases are reduced modulo the operand, with zero residues skipped. The word backend uses 128-bit modular products. For larger operands, the program uses the first twelve prime bases, $2, 3, \ldots, 37$. Sorenson and Webster establish that the least composite passing all those tests is

$$\psi_{12} = 318665857834031151167461 \tag{18}$$

[19, Theorem 1.1 of the preprint]. Only x_7 among the factored operands can exceed 64 bits; it is at most $U_7 < \psi_{12}$. The wide test is guarded strictly below ψ_{12} itself. Its modular multiplication and exponentiation use GMP; native 128-bit products would not cover every product of two 67-bit residues. The producer uses the word helper whenever the operand fits and dispatches to the wide helper only when needed.

These finite-domain premises are cited external computational results, not new exhaustive primality computations performed in this study. The negative census retains them as dependencies. Trial division of the reported positive chains does not use these premises.

5.3 Parallel ownership, coverage and recovery

The first-derivative sieve is followed by an OpenMP loop with dynamic chunks of 64 candidates. Each worker owns its GMP scratch and writes a separate result flag. The implicit join precedes

the serial examination of flags and replay of positive inputs in increasing order [22]. A block is committed only if all its candidates have completed. The program's statuses distinguish positive, negative and incomplete evaluations. A signal or resource failure prevents advancement past the incomplete block.

To specify the partition, let $w = 262144$ and let $K = \lceil (H-L+1)/w \rceil$ be the number of blocks in a configured interval. With σ operational segments, segment i, $0 \le i < \sigma$, uses block indices $\lfloor iK/\sigma \rfloor$ through $\lfloor (i+1)K/\sigma \rfloor - 1$. Clamping the last upper endpoint to H gives a disjoint consecutive partition. The state records configuration identity, segment bounds, completed frontier, block count, saved work time, fallback count, and ordered positive tuples. Every completed segment receives a seal. Consolidated manifests and positive tables are checked against those authoritative states; a future unstarted segment records no examined interval. Integrity records detect changed bytes and bind the recorded producer; they do not by themselves prove that each negative classification was mathematically correct.

6 The finite result and its validation

Theorem 6.1 (Computational extension, with explicit dependencies)**.** *Subject to the accepted five-term baseline, correct execution of the AI code-reviewed enumeration described in Section 5, its stated numerical and finite-domain primality premises, and the complete saved records, the only members of S_8 in the inclusive interval* $823544 \le n \le 22000000000$ *are*

$$1072281223, \quad 1320016916, \quad 8910114183, \quad 10361580899.$$

They have consecutive ranks six through nine. No further member occurs after $a(9)$ through the inclusive endpoint 22000000000*.*

Proof (computational). The classified intervals are displayed in Table 4. The lower endpoint of each continuation is exactly one greater than the preceding upper endpoint. There are 34 sealed segments and 83921 committed blocks. Their input counts sum to

$$1071457680 + 19174260736 + 1753458041 = 21999176457 = 22000000000 - 823543.$$

The complete-block rule and checked partitions supply the recorded coverage, under the producer assumptions in the statement. The first interval has only N, exactly at its upper endpoint. The second has the other three positives in increasing order and the third has none. Their full chains agree with the documentary table. Appending these four hits to the inherited prefix assigns the displayed ranks. Independent positive reconstruction checks the eight derivative identities and square identity for every row; the exclusion component rests on the producer and records, not on that positive-only check. □

Portion used	Inclusive lower	Inclusive upper	Seals	Hits
First run	823544	1072281223	2	1
Continuation prefix	1072281224	20246541959	29	3
Final run	20246541960	22000000000	3	0

Table 4: Executed coverage after the inherited five-term prefix. The continuation was configured for 50000000000 and remains incomplete relative to that original target. Only its sealed 29-segment prefix contributes. The final run is separately complete.

Corollary 6.2. *With the computational dependencies of Theorem 6.1,*

$$22000000000 < a(10) \leq 285311670611 = 11^{11}. \tag{19}$$

The value of $a(10)$ *is unknown.*

Proof. The theorem gives the strict lower bound. The known stationary family supplies $11^{11} \in S_8$ above the frontier. Infinitely many prime parameters supply infinitely many distinct members, so the next rank exists. The witness 11^{11} need not have rank ten; it supplies only the upper bound. □

The original continuation configuration was preserved when the author selected the narrower endpoint. The final adjacent run has its own configuration; no old target or unstarted state was relabelled as completed coverage. The supported input cap remains $5{\cdot}10^{10}$, which also lies below the mathematical upper witness. Capacity is not evidence that those additional inputs were examined.

6.1 What the independent checks establish

The independent Python record checker imports no producer code and invokes no producer. It checks configuration and state grammar, frontiers, seals, ordered joins, counts and correspondence with the consolidated records. For every positive tuple, it independently factors by complete trial division, reconstructs the chain, and uses integer square roots. The documentary-table checker compares the indexed and witness tables row by row, checking all 72 derivative transitions and all nine square identities. Python arbitrary-size integers and `math.isqrt` [23] give an arithmetic path distinct from the C factorization and GMP square test.

These checks confirm saved positive arithmetic and record consistency. They do not independently visit every negative input in the added interval, and they share the mathematical definition and saved inputs with the producer. Multiple Python implementations of this arithmetic are corroboration of the same positive axis, not multiple exhaustive scans. The original first five ranks and their gaps remain inherited, even though their membership is rechecked. The retained finite checker for this paper additionally normalizes the nine tuples, checks every generalized transition, reconstructs the majorant constants and manifest totals, and tests the stated depth correlations on these nine inputs.

The unchanged development evidence includes small direct comparison windows, known positives, boundary/refusal cases, forced fallback, signal/recovery and parser tests, and word/wide arithmetic fixtures. Some historical development drivers require the original development tree and dependencies and are not replay commands for the review extract; the independent record, documentary-table and finite structural checkers described above are replayable. The wide fixtures include recursive Lucas $n-1$ prime certificates with checked factor products; these are scoped checks of selected operands. A maximum-width transport fixture was deliberately not a mathematical witness and was rejected by the mathematical verifiers. It contributes no sequence term. Recorded GCC/Clang, sanitizer, static-analysis and qualified OpenMP race-detector checks support the implementation obligations in their tested domains; none is a mathematical certificate for all negatives.

6.2 Recorded environment and measurement scope

The frozen environment records an AMD Ryzen 9 7940HS with eight physical cores and sixteen logical processors, Linux Mint 22.3, kernel 6.17.0-42, GCC 13.3.0, GMP 6.3.0 and OpenSSL 3.0.13. The runs use sixteen workers and record OpenMP macro value 201511. The build uses `-O3`

`-std=c17 -fopenmp`, strict warning options promoted to errors, and links GMP and libcrypto, without processor-specific tuning flags.

The sum of saved completed-block work counters is approximately 26595 seconds, about 7 hours 23 minutes: approximately 914, 23332, and 2349 seconds for the three portions respectively. These counters time completed block work, not complete campaign wall time. They exclude checkpoint and audit overhead, pauses, and any repeated unfinished work. The saved fallback-input count is zero; that observation does not remove the fallback contract or bound the cost of future inputs. No sustained campaign peak RSS, installed-memory capacity or thermal qualification is inferred from short development measurements.

6.3 Local reproducibility and publication boundary

The local project retains the source and build, original executable, canonical indexed and documentary tables, configurations, sealed states, manifests and runnable Python validators. Auditing the saved records and reconstructing the positive witnesses requires no new official enumeration. A complete public reproduction also requires those actual run payloads and their configuration bindings; the manuscript's tables alone cannot reproduce the negative census. The project reference [25] identifies the public repository and Zenodo release series. The initial source-and-evidence package is available as a GitHub release and as an open Zenodo deposit. The accompanying package provides runnable retained-data, positive-witness and structural checks. Those checks audit the saved evidence; they do not independently repeat the full negative census.

7 Consequences and open questions

The finite prefix combines four stationary examples, two members of one nonconstant infinite form, and three singleton forms. Thus two different distinctions are necessary: membership versus rank, and complete scales of a fixed form versus a classification of all forms. A new family member gives an upper witness, while its rank requires intervening exclusions. Likewise, the rate criterion completely determines the scales of F_1 but supplies no complete enumeration of other primitive shapes. Any further member strictly below 11^{11} must have a primitive form absent from Table 1: the next F_0 member is 11^{11}, the next F_1 member is larger, and the three E_i classes have no further scales. This does not force $a(10)$ to have a new form, since equality with the upper witness remains possible.

The general rate theory permits rational rates, subject to Theorem 2.4. All five observed forms have rates zero or one. Whether an admissible depth-eight form has a noninteger rate, whether infinitely many primitive forms occur, and whether infinitely many of them are singleton or terminating forms remain unresolved here. So does the least member above the completed frontier. The nine observations provide no justified density law, probability model for a future hit, or asymptotic estimate. The two infinite classes and the three specified singleton classes, in contrast, follow by proof rather than extrapolation.

8 Conclusion

The completed added coverage extends the accepted five-term prefix to nine consecutive terms and excludes further members through 22000000000, with explicit production and baseline dependencies. Its first added rank belongs to a known persistent zero-tail witness. The ordered forms separate two infinite homothetic classes from three singleton classes, and the nonconstant class

explains the proportional tuples at ranks seven and eight. These structural results account for the observed prefix without classifying all possible forms. The next rank is bounded by a historical fixed-point witness and requires further minimality evidence for its determination.

Data availability

The source and build, original configurations and sealed records, indexed b-file, documentary derivative/root table, independent positive and record checkers, finite structural checks, environment captures and validation summaries are available from the public GitHub repository and Zenodo release series cited in the References. The package also includes this monolithic LaTeX source and its matching PDF. The repository documents the verification commands, dependencies and restoration of the original evidence layout. The Zenodo DOI identifies the release series, rather than a single immutable version; individual deposits retain their own version-specific DOIs. Positive reconstruction and record audits do not independently certify all negatives in the reported interval.

AI use disclosure

OpenAI Codex assisted with research, implementation work, local verification and preparation of this manuscript. The supplied code review reports attest reported contributions from Anthropic Claude, GLM/Z.ai, xAI Grok, Meta AI, MiniMax and Qwen, alongside OpenAI. These are reported families rather than independently authenticated backend identities; exact versions are not consistently established. A subsequent manuscript round supplied ten AI-attributed reports. Their comments were adjudicated against the manuscript and primary sources; model identities and reported executions were assessed separately from the evidence actually supplied.

Mathematical analysis. Codex assisted in reconciling the OEIS definition and historical witnesses, developing the product-rule and fixed-form arguments, checking the nine normalized tuples and investigating depth relations. The mathematical statements are supported by the displayed proofs, primary source readings and exact finite calculations, with historical principles attributed to their sources.

Algorithm and code. Codex assisted with the C implementation and its numerical, parallel and recovery contracts. The reported external families above contributed code review; their reports were adjudicated against the source and retained tests. The author executed the official campaigns. Codex executed the retained Python record and positive checks locally during manuscript preparation and revision. The numerical result is tied to that producer's records, known-term checks, separate positive arithmetic and scoped development tests. Code-review agreement is not an independent negative enumeration or manuscript review.

Manuscript. Codex assisted in English and LaTeX drafting, literature checking and internal review of proofs, numerical statements, chronology, citations and availability. Final numerical claims rely on local artifacts and validation, not conversational assertions. The new manuscript proofs and the adjudicated review findings remain subject to the author's scientific assessment. The recorded checker executions identify their operator; they do not attest to a personal rerun of every check by the author. Carlo Corti retains full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries and conclusions.

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